A Host Mobility Support with Adaptive Network Selection Method in Hybrid Wireless Environment

Mohammad Reza Heidarinezhad[*Corresponding author], *Zuriati Ahmad Zukarnain*[*],
Nur Izura Udzir [**], *Mohamed Othman*[*]
[*]*Dept. of Communication Technology and Network,*
[**]*Dept. of Computer System*
Faculty of Computer Science and Information Technology, University Putra Malaysia
{heidari, zuriati, izura, mothman}@fsktm.upm.edu.my
doi: 10.4156/jdcta.vol3.issue1.heidarinezhad

Abstract

The B3G or 4G of cellular/wireless communications network is expected to be purely IP-based and consist of heterogeneous access networks from 3G cellular, WiFi, WiMAX and a converged core network. To support the mobile host in the hybrid overlay wireless, a cross-layer vertical handoff control method has been proposed. Implemented architecture is composed of four part; Connection Profile Manager (CPM), Network Access Assistance (NAA), Neuro-Fuzzy Decision Engine (NFDE), and Peer-bind Connection Manager (PCM). The most effective connection is selected based on the attributes from the multi-layer platform that provided handoff decision component which is responsible for handoff decision. The proposed network selection algorithm is based on hybrid neuro-fuzzy concept with low packet loss and latency. We have analyzed our implemented model on different scenarios. Results of the experiment indicate the advantages of the proposed scheme.

Keywords

Handoff, mobility management, neuro-fuzzy, wireless network

1. Introduction

It is avouched that next generation of wireless networks will utilize different access technologies such as WLAN(802.11x), WWANs such as General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), Code division multiple access (CDMA) and WiMAX (World Wide Interoperability for Microwave Access) and other emerging access technologies such as mesh networks. The main aim of cooperation of diverse wireless technologies is providing ubiquitous access to demanded services. Each one of above mentioned technology has own specification in term of QoS level, coverage area, bandwidth and cost. The incoming mobile terminals (MTs) such as smart phones and PDAs, will capable of multimode access interface to supporting different types of radio access technologies on single equipment.

Beside, with rapid growing of wireless infrastructure technology, the areas with multiple coverage of the umbrella-like wireless access are increased. Hence supports of nomadic users to enjoy seamless connection, have been focused by many researchers in academia and industrial. In heterogeneous data networks, an accurate decision on handoff has a significant impact on the throughput of the system. More precisely, one of the main issues of the handoff management is concerned with deciding the best attach point while a MT is in the coverage area of multiple diverse networks that been shown in Figure 1. In zone A and B, mobile host locate in three and two candidate access network, respectively. Besides, the traditional handoff mechanism, actually was based on received signal strength (RSS) or received power (P) which are not usable for heterogeneous wireless environments due to diversity of physical links. The above mentioned issues are the main motivation behind of this works. Also propose efficient handoff decision scheme with consideration of user preference, QoS of current service and available networks conditions can assumed as objective of this studies.

The following issues are important design goals for our cross-layer mobility solution:

1- Small handoff latency and efficient data packet delivery.
2- Ease of administration and deployment.
3- Scalability and robustness.
4- Application transparency.
5- User preferences and service cost.

These objectives can be archived mainly through fast decision making in NFDE module as well as peer-to-peer packet forwarding in CPM, NAA and PCM component.

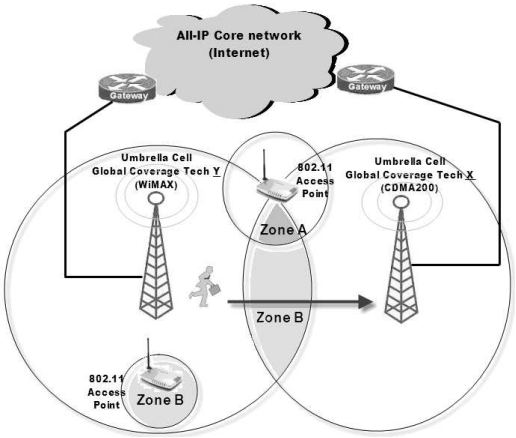


Figure 1. Multi-coverage zones in hybrid wireless environment

In this article, we present a combined neuro-fuzzy handoff decision method as a major component of handoff control system. The main advantage of using neuro-fuzzy genre in optimal selection modeling is that they can be synthesized without making use of the detailed, explicit knowledge of the underlying process. Because of their complementary nature, these two technologies can be integrated in a number of ways to overcome the drawbacks of each other. Our experiments shown that performance of network selection procedure noteworthy depend on various dynamic parameters such as subscriber motion model, number of training samples and traffic distribution.

The rest of this paper is organized as follows. Section 2 reports a review of previous related works especially for network selection algorithms. In section 3, we present architecture of proposed system, include the sketch of NFDM and detail of implemented system. In section 4 the result of experiments has been described. The conclusion is discussed in section 5.

2. Related Work

To support MT across hybrid wireless networks, many mobility management schemes have been proposed. Mobile IP (MIPv4 and MIPv6) [1,2] and its extension is the most well-known solution investigates mobility support at L3. Beside, a further solution for maintenance the connection dealing at the application layer, based on end-to-end (E2E) method is defined in Ref. [3]. In this scheme after changing the IP address of MT, it modifies the communication peer directly.

Beside, vertical handoff decision has an important role in effectiveness on host mobility. Two main categories of vertical handoff decision scheme are proposed in the prior literatures: *function-based* and *mathematical-based*. In function-based mechanisms, the best attach point is chosen based on output of given functions. Objective function, consumer surplus, profit function and degradation utility are existing instance in this category. Beside this, mathematical-based category emphasis on multi-criteria based methods such as artificial intelligence, pattern recognition and genetic algorithms. Recent category in addition to RSS, combines several metrics such as bandwidth and service cost for handoff decision. Fuzzy logic-based, stochastic linear-based, game theory and AHP-GRA-based approaches fall in mathematic-based category. Ref. [4] is an excellent survey of decision management techniques. For example Ref. [5] is being used a neural networks-based approach to detect signal decay and making handoff decision across GPRS and WLAN. As another attempt, in Ref. [6], SAVA is proposed as a vertical handoff decision method using the self-adaptive approach.

The best network selection in vertical handoff based on neuro-fuzzy concept have been proposed recently in literature in many different fields such as radio resource management [7,8]. Nevertheless, this type of algorithms has already been proposed by the authors in their previous works, which mainly focuses on the implementation of a novel network joint algorithm [9].

3. Proposed Scheme

This section discusses design for handoff decision making procedure and handoff execution. Cause of complexity and the multiple dynamic aspects of host roaming, a design based on a cross-layering architecture interleaved with correlative components is almost an unavoidable choice to achieve adaptive behavior and performance optimization.

Implemented architecture composed of four parts: Connection Profile Manager (CPM), Network Access Assistance (NAA), Neuro-Fuzzy Decision Engine (NFDE), and Peer-bind Connection Manager (PCM). Figure 2 depict the block diagram and correlation of components for proposed scheme.

The decision making for the optimum available network is choosing by NAA. The neural computing and fuzzy systems are suitable to deal with complex problems efficiently. These methods recently used for improve handoff trigger time, separately. We believe a

synergy of fuzzy logic and artificial neural networks can improves efficiency, performance, and adaptiveness of yields.

The MT should support multi mode that can detect different frequencies. In this paper, we don't consider Authentication, Authorization, and Accounting (AAA) between different network providers. For location management issue, we supposed that when a MT is attached to one served network, destination infrastructure is responsible for that function.

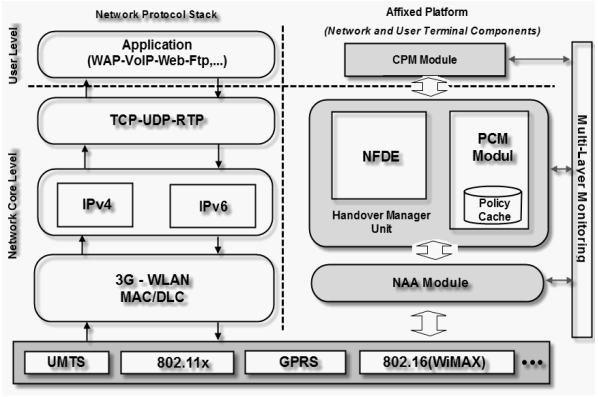


Figure 2. Architecture of proposed multilayer scheme

3.1 CPM

Connection Profile Manager keeps user preference settings for handoff execution. To meet the basic concept that different type of application need different kind of mobility support, CMP performed. The main reason for engender connection profile is enable the user to easily choice her/him pre-exist setting whereas MT enter to new coverage area. Therefore, user can switch between profiles to carry out proactive handoff policy. For example, a typical user may desire to organize two type of setting on his/her equipment. First one for QoS sensitive application such as real-time services and VoIP to support seamless handoff during attach to new point of access although with more cost. On the contrary, the second setting may for mass data communicate such as web surf and file download/ftp with less charge. CMP should be implemented as user application daemon. This module provides a very simple GUI for user to create and modify profiles with desired variable parameter value for different connection scenarios such as access cost. Because this component actually employed in the user equipment, our approach falls in the mobile assisted handoff management category.

3.2 NAA

Network access assistance collects RSS level from different network interface and reports them to NFDM module. Due to use of multiple radio access interfaces, this module also specifies use of physical interface before/after handoff. RSS values, which gathered from lower layer, will be used to feed in the NFDE function.

3.3 PCM

In order to maintain the continuity current session, Peer-bind Connection Manager module provide a according to peer-to-peer (P2P) technology. Handoff management scheme dealing with P2P principle such as [10, 11,12] has been shown superior performance to support mobile nodes. After the handoff is triggered, mobility scheme have to keep network connectivity to impart continue data packet flow. PCM also contains a policy cache that is a repository to store connection profile manager in system side. However, an MT may not have a fixed and previously known IP address. Consequently, PCM allows Corresponding Node (CN) and MT to add, delete or change IP addresses during an active SCTP association using address configuration (ASCONF) messages. The SCTP-based vertical handoff approach uses the multi-homing feature of SCTP so that a mobile client can have two IP addresses during the vertical handoff, one from the UMTS and the other from the WLAN. Both, the MN and the fixed server need to implement mSCTP. The MN also needs to support both, UMTS, WiMAX and WLAN at the physical and data link layers. The handoff procedure across hybrid environment is assumed as mSCTP protocol in the next step.

3.4 NFDE

Decision making for select optimal network is a multidimensional, uncertain and approximate reasoning problem that is an appropriate candidate for resolve based on neuro-fuzzy method. NFDE actually is developed based on Adaptive Fuzzy Logic System (AFLS) [13]. The AFLS is one type of fuzzy logic system (FLS) with fuzzifier and defuzzifier units. Its structure is the same as a conventional FLS but its rules are derived and extracted from given training data. In other words, its parameters can be trained like a neural network approach, but with its structure in a fuzzy logic system structure. The centroid defuzzifier is the most eminent method for defuzzification but cannot be used in NFDE due to its computation expense and prohibits using of back-propagation training algorithm. The proposed AFLS includes an alternative defuzzification

approach based on FALCON model [14] presented by Altug. The feed-forward NFDE structure according to FALCON model is shown in Figure 4.

NFDE is triggered when 1- The mobile host detects the new radio link; 2- Changing user preference 3- a new service request is made; and 4- There is severe signal degradation or complete signal loss of the current radio link.

Membership Functions Definition

For the best network selection in vertical handoff, employ suitable parameter from different layer both the user side and the system are required. Attributes that used for the NFDE include monetary cost (C), Network Bandwidth (B), RSS (R), User preference (U) and network delay (D). Impart better control to result is possible via increase number of fuzzy sets but this issue will amplify complexity of the NFDE. Membership functions must be adaptive to the changing environment in order to continue to be useful. System behavior significantly is reliant on fuzzy rules and membership functions to describe the decision rules. The NFDE's attributes are defined as:

C= {Economic, Normal, Expensive}
B= {Poor, Med, Good} //Mbps; MTS=Poor; WLAN = Good
R= {Low, Normal, High} //dBm
U= {Low, Normal, High} //Sensitive to QoS=High;
Ordinary usage=Normal, Insensitive = Low
D= {Low, Med, High}

Figure 3 shows the membership function of the input fuzzy attributes for five defined variable.

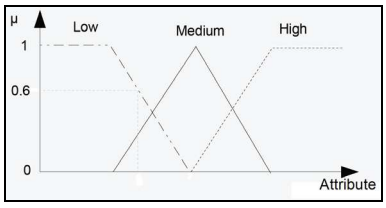


Figure 3. Membership functions of the input variables

Each fuzzy input variable deal with a number of If-Then fuzzy rules that can be shown as:

IF C is Economic AND B is Med AND R is Normal AND U is High AND D is Low THEN handoff factor = Out(C,B,R,U,D).

Where *Out*(.) is output of normalize model that fed into the next layer. According to FALCON model, the final output of neuro-fuzzy system determine as equation(1):

$$Y = F(C,B,R,U,D) = \frac{(W^{ij})Out(.)}{\sum O(x)^i} * \frac{\sum((W^{ij} * O(x)^i)}{\sum w^{ij}} \quad (1)$$

Where W^{ij} is the weight of i-th neuron in layer j and $O(x)^i$ is the output of normalized function of i-th input value ($0 \le O(x^i) \le 1$). The selected network has the highest Y. The gain threshold is considered as (0.8 * $Y_{Current\ Service}$) for Y. Future detail of this neuro-fuzzy architecture and their functions are outlined in [14]. The suggested architecture for such neuro-fuzzy based approach has been shown in Figure 4.

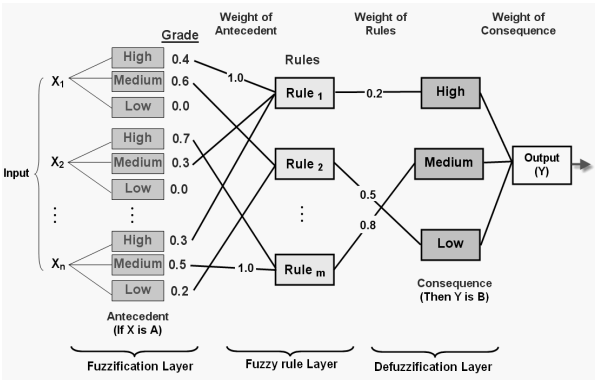


Figure 4. Architecture of neuro-fuzzy-based decision maker in NFDE

The flowchart of the wireless link selection based on cross-layering criteria in NFDE is reported in Figure 5. X(i) in this procedure indicate the {C,B,R,U,D} fuzzy set. Fuzzy attributes records by MT and in additional with user preference and weigh of neurons will be store at CPM's profile.

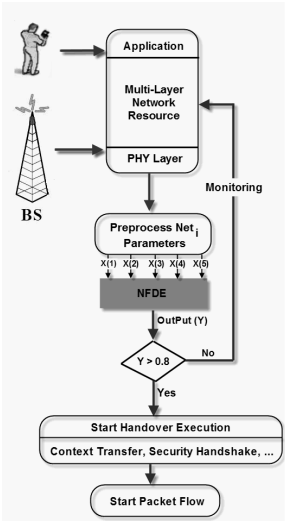


Figure 5. Wireless network selection procedure

4. Experimental Results

To evaluate the proposed network selection algorithm, we supposed behavior of a typical user during a number of days from source to destination (e.g. home to office).With obtained training data; the *W* matrix will be formed. For further realities, user request categorized into three different services. The redefined value for data set{C, B, R, U, D} to perform operation in NFDE are indicated in table 1.

Table 1. Value of fuzzy parameters for case study

	C (Cent/Kb)	B (Mbit/s)	R (dBm)	U (U.P)	D (ms)
WLAN	0.001	11	+38	0.90	1.25
WiMAX	0.185	2	+30	0.50	12.25
UMTS	0.220	0.5	-100	0.15	18.54

Simulation environment includes MT, HA, CH and foreign access router.

The simulate scenario also include a two-tier cellular network (i.e. WiMAX and UMTS cell overlays) and a WLAN access point. The experimental setup has been shown in figure 6.

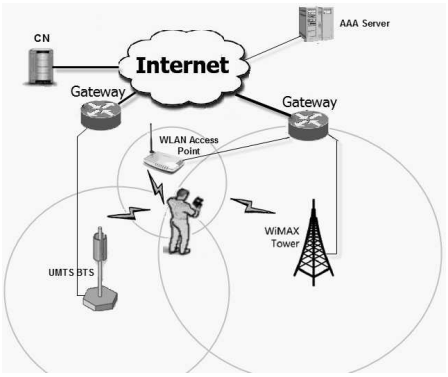


Figure 6. Experimental Environment

As it can be seen, the MT can connect to each available network and continue current service. We also assumed the MT is equipment with multimode wireless interface. To observe response times, we have chosen three various services include FTP, web browsing and VoIP, with different QoS requirement. The bandwidth offered by the WLAN, UMTS and WiMAX is considered 11Mbit/s, 0.5Mbit/s and 2Mbit/s, respectively. Because line-of-sight considered same for all access points, SIR, SNR and velocity of mobile node did not deemed in simulation. After the training step, the *W* matrix for our case study is indicated as below:

$$W = \begin{bmatrix} 0.0652 \\ 0.0791 \\ 0.0211 \\ 0.0981 \\ 0.4991 \end{bmatrix}$$

Furthermore, Table 2 shows the response time while handoff between three access points with diverse services. As the model of movement of MT is changed in the subnets, the handoff decision delay and finally response time noteworthy is increased, leading to an increase of the packet loss rate.

Table 2. Response times (sec) for roaming across UMTS-WLAN-WiMAX

Network	Application type	FTP File download	Web browsing	VoIP
WLAN	Min	0.775	0.011	0.00044
	Max	3.675	0.055	0.00055
	Ave	1.470	0.029	0.0005
UMTS	Min	6.944	0.900	0.344
	Max	24.690	1.699	0.350
	Ave	14.721	1.191	0.347
WiMAX	Min	2.325	0.154	0.0042
	Max	8.321	0.876	0.0087
	Ave	6.543	0.468	0.0066

In Figure 7, we show the distribution of forecast miss rate of each MT. In the same scenario, with using our scheme, we forecast the best connection for more than one available link. With forecast of best link according to the *W* case study matrix, about 82% of the MTs have a forecast miss rate less than 0.5. With forecast of one subnet, about 82% of the MTs have a forecast miss rate of less than 0.45. Compared with the miss rate of randomly selecting two or more available access points, our proposed scheme has a much lower miss rate. Only about 7% of the MTs have a miss rate less than 0.3 when randomly selecting available existing access points. We also has been compared our method with neuro-mathod as shown in Figure 7.

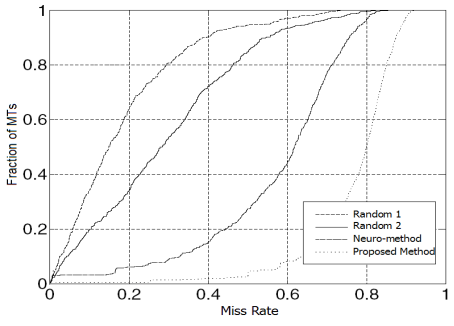


Figure 7. Comparison of response time between three wireless technologies

Its miss rate is shown by the middlemost line in the Figure 7, and is greater than zero because a miss is incurred when a movement from one specific access point to another occurs for the first time. This confirms the value of predicting MTs' movement based on history.

5. Conclusion and Discussion

We propose a cross-layer host mobility support with adaptive handoff decision based on neuro-fuzzy concept. Handoff decision procedure determines whether a vertical handoff should be executed. The proposed scheme dynamically chooses the optimum connection from available access network technologies to continue with an existing service or begin a new connection. To archive seamless handoff with user satisfactory result, connection profile from application layer in addition to L2 parameter has been used.

The most important gain of proposed approach is to minimize modification of network protocol stack due to using standard mSCTP protocol as handoff solution. Furthermore, our method does not require any amendments to existing infrastructure. We prove the feasibility of our proposal by a performance evaluation. The results of experiments indicate the practicability of proposal.

The major drawback of neuro-fuzzy based methods affiliate to find optimum weight of neurons and appropriate normalization fuzzy rules. For example, limited or noisy training data may result in inconsistent, meaningless outputs.

To contemplate more realistic circumstances, may need to consider further situation such as battery status, traffic model of serving BS to find suitable candidate network to switching. In addition, there are number of QoS factors such as traffic model that have significant impact on consequence. Further work may establish dynamic causal relationships between metrics and criteria may be satisfied with realistic mobility frameworks.

6. References

[1] C. Perkins, IP mobility support, in IETF RFC 3344, August 2002.

[2] D. Johnson, C. Perkins, and J. Arkko, "Mobility Support in IPv6", RFC 3775, IETF Network Working Group, June 2004.

[3] G. A.Di Caro, S. Giordano, M. Kulig, D. Lenzarini, A. Puiatti, and F. Schwitter, "A Cross-Layering and Autonomic Approach to Optimized Seamless Handover", Proceedings of the 3rd Annual Conference on Wireless On demand Network Systems and Services(WONS), 2006.

[4] Ksentini, Kpcva and J. M. Bonnin, "PI-1889: Resource Management in Mobile Heterogeneous Networks: State of the Art and Challenges", IRISA research reports, Feb 2008.

[5] K. Pahlavan, P. Krishnamurthy, A. Hatami, M. Ylianttila, J.R. Makela, and J. Vallstrom Pichna, "Handoff in hybrid mobile data networks", IEEE Personal Communication, Vol. 7, pp. 34–47, Apr. 2000.

[6] Liu Min, Li Zhongcheng, Xiao-Bing Guo, Eryk Dutkiewicz, and Ming-Hui Wang, "SAVA: A Novel Self-Adaptive Vertical Handoff Algorithm for Heterogeneous Wireless Networks", IEEE Global Communications Conference 2006 (GLOBECOM 2006), San Francisco, USA, 2006.

[7] K.R. Lo and C.B. Shung, "A Neural Fuzzy Resources manager for Hierarchical Cellular Systems Supporting Multimedia Services", IEEE Transactions on Vehicular Technology, Vol. 52, No.5, pp. 1196-1206, 2003.

[8] L. Giupponi, R. Agustí, J. Pérez-Romero, and O. Sallent, "Joint Radio Resource Management Algorithm for Multi-RAT Networks" , GLOBECOM-2005, St Louis 28, 2005.

[9] Liu Xia, Ling-Ge Jiang and Chen He, "Novel Fuzzy Logic Vertical Handoff Algorithm with Aid of Differential Prediction and Pre-Decision Method", IEEE International Conference on ICC apos-07, No.24, pp. 5665-5670, 2007.

[10] C. Guo, Z. Guo, Q. Zhang, and W. Zhu, "A seamless and proactive end-to-end mobility solution for roaming across heterogeneous wireless networks", Selected Areas in Communications Special Issue on Advanced Mobility Management And QoS Protocols for Next generation Wireless Internet, Vol. 22, No.5, pp. 2004.

[11] I. Stoica, R. Morris, D. Karger, M. Kaashoek, and H. Balakrishnan, "Chord: A scalable peer-to-peer lookup service for Internet applications", in Proceeding of SIGCOMM, San Diego, CA, pp. 149–160, 2001.

[12] Chuanxiong Guo, Zihua Guo, Qian Zhang, and Wenwu Zhu, "A Seamless and Proactive End-to-End Mobility Solution for Roaming Across Heterogeneous Wireless Networks", IEEE journal on selected areas in communications, Vol. 22, No.5, pp. 834-847, June 2004.

[13] L.X. Wang, Adaptive Fuzzy Systems and Control, Prentice Hall, Inc, 1994.

[14] S. Altug, J. Trussell, And M.Y Chow, "A mutual update training algorithm for fuzzy adaptive logic control/decision network (FALCON)", EEE transactions on neural networks, Vol. 10, No.1, pp. 196-199, 1999.